CHAROTAR UNIVERSITY OF SCIENCE &TECHNOLOGY

Fifth Semester of B.Tech Examination (IT)

November-December 2015

IT302.02: Design and Analysis of Algorithm

Date: 28.11.2015,Saturday **Time: 10:00 a.m to 1:00 p.m** **Maximum Marks:70**

Instructions:
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1(a) **Answer the following questions.**

1 Consider the following functions: **01**

$f(n) = 2^n$

$g(n) = n!$

$h(n) = n^{\log n}$

Which of the following statements about the asymptotic behavior of f (n), g (n), and h (n) is true?

(A) $f(n) = O(g(n)); g(n) = O(h(n))$

(B) $f(n) = \Omega(g(n)); g(n) = O(h(n))$

(C) $g(n) = O(f(n)); h(n) = O(f(n))$

(D) $h(n) = O(f(n)); g(n) = \Omega(f(n))$

a) A b) B c) C d) D

2 If f (n) = O (g (n)) than **01**

a) g(n) is asymptotic lower bound for f(n)
b) g(n) is asymptotic tight bound for f(n)
c) g(n) is asymptotic upper bound for f(n)
d) None of these

3 What does it mean when we say that an algorithm X is asymptotically more efficient than Y? **01**

a) X will be a better choice for all inputs
b) X will be a better choice for all inputs except small inputs
c) X will be a better choice for all inputs except large inputs
d) Y will be a better choice for small inputs

4 Let f (n) = n and g (n) = $n^{(1+\sin(n))}$ where n is a positive integer. Which of the following statements is/are correct? **01**

I. $f(n)=O(g(n))$
II. $f(n)=\Omega(g(n))$

a) Only I
b) Only II
c) Both I and II
d) Neither I nor II

5	NP-Complete ∩ P = _____which of the following is true?	01
	a) NP-Hard b) ϕ c) P d) NP-Complete	
6	Define P, NP, NP complete and NP-Hard problems. Draw a relationship diagram between P, NP, NP Hard and NP complete	02

Q-2 **14**

(a) Define Time Complexity with suitable example. Arrange the following terms in increasing order of their time complexity for large values of n. **04**

$O(n^{(3/2)})$, $O(n)$, $O(1)$, $O(\sqrt{n})$, $O(n!)$, $O(\log\log n)$, $O(2^n)$, $O(n\log n)$

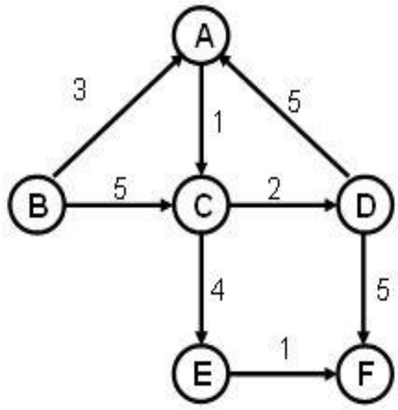
(b) Answer Any Two Questions. **10**

1 What is Asymptotic Notation? Explain O, Ω, θ notation with a suitable example and neat sketch diagram.

2 Let S= {a, b, c, d, e} be a collection of objects with benefit weight values as follows: What is an optimal solution to the **fractional knapsack** problem for S assuming that knapsack can hold objects with total weight 35 using greedy approach?

Item	Value(P)	Weight(W)
1	20	10
2	35	14
3	25	12
4	40	7
5	30	15

3 Execute the Dijkstra's algorithm on the following graph, with single-source ***B***

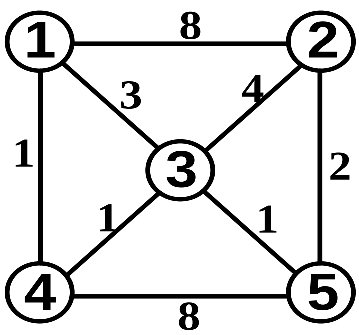


Q-3 **Answer Any Two Questions**. **14**

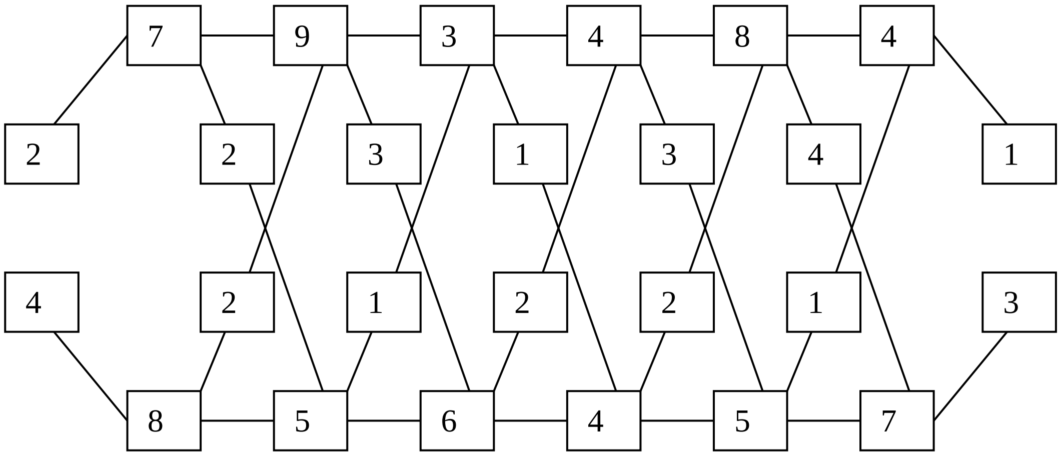
(a) Find Upper bound of following recurrence equation using Recurrence Tree Method.

1. $T(n) = 2T(n/2) + c\, n^2$
2. $T(n) = 3T(n/4) + cn^2$

(b) A weighted graph is given as below. Find all pair shortest path using Floyd Algorithm.



(c) A car assembly plant having two assembly lines with six stations on each the time required by each station is given below in figure.



If entry & exit time for line 1 is 2, 1 and for line 2 is 4, 3 then find optimal path & time by which car can be moved within assembly plant.

SECTION – II

Q-4 **Answer the following questions.**

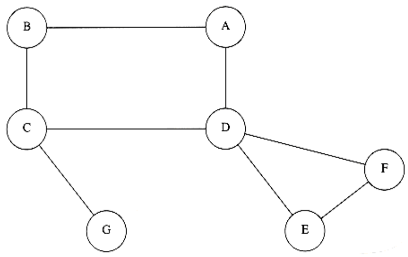
1 To find minimum spanning tree of a given graph, Kaushal's Algorithm is preferred when, **01**

a) Given graph is Dense
b) Given graph is Sparse
c) Given graph is not connected
d) None of the above

2 Which among the following is the best when the list is already sorted **01**

a) Insertion sort
b) Simple Bubble sort
c) Merge sort
d) Selection sort

3 Total number of articulation point in the following graph is __________. **01**



a) 0 b) 1 c) 2 d) 3

4 What is feasible solution? Write two feasible solutions for 4-Queen problem. **02**

5 Greedy Approach always gives feasible solution not optimal. Justify your answer **02**

Q-5

(a) Differentiate Greedy approach and Dynamic programming. Explain with an example of making change problem. **04**

(b) **Answer Any Two Questions**. **10**

1 Assigning 4 people to 4 jobs so that the total cost is minimized. Each person does one job and each job is assigned to one person. Use branch and bound method to solve the assignment problem (Minimization) with the following cost matrices.

	Job 1	Job 2	Job 3	Job 4
Person a	9	2	7	8
Person b	6	4	3	7
Person c	5	8	1	8
Person d	7	6	9	4

2 What is Primitive Operations? Find total no primitive operations and growth rate of a given algorithm.

```
Algorithm Fibonacci(n)
{
if ( n <=1) then
            Write(n)
      else
            fnm2=0;
            fnm1=1;
            i=2;
            for i ← 2 to n do
                  fn=fnm1 + fnm2;
                  fnm2=fnm1;
                  fnm1=fn;
            write (fn)
}
```

3 How many exact match and spurious hits does the Rabin-Karp matcher encounter in the text T = 564159265359 when looking for pattern P = 26 Take value of modulo q= 11.

Q-6 **Answer Any Two Questions**. **14**

(a) Write General recurrence equation for divide and conquer technique. Write recurrence relation for Worst and Best Case of Quick Sort and Using recurrence relation determine asymptotic growth rate in term of Big O notation for Worst Case and Best Case.

(b) State Master Theorem and Solve following recurrence using Master Theorem.

1. $T(n) = 7T(n/3) + n^2$
2. $T(n) = 3T(n/3) + \sqrt{n}$

(c) There are eleven Professors in a Department. Each professor wants to deliver lecture in same day. Each professor has some time limits for lecture. Professor earns credit if and only if lecture is arranged on or before its time limit.

Professor	1	2	3	4	5	6	7	8	9	10	11
credit	78	90	50	60	75	10	80	55	88	74	59
Lecture Limit	5	4	5	3	2	1	4	6	4	5	6

Maximize the total credit earned by all professors of the Department using greedy approach. What is the time complexity of your algorithm?

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